

COP 3515 – Advanced Program Design

Semester Programming Project

Client Change Request (CCR-003)

Project: Student Information Management System (SIMS)

Customer: University Registrar's Office

Priority: High

Assigned To: Student Developer

Sprint: Week 3

Due Date: Beginning of Lesson 8

Business Background

The University Registrar's Office has successfully implemented the first two releases of the Student Information Management System (SIMS). The system now records student information and course grades, allowing academic advisors to review a student's current performance.

The Registrar would now like the system to determine each student's **academic standing** automatically. Currently, advisors manually review each student's GPA and classify students into various academic categories. This process is time-consuming and occasionally results in inconsistent classifications.

The Registrar would like the software to classify students into one of several academic standing categories based on their GPA. The system should also reject GPAs that are not valid.

This enhancement should be added to the existing SIMS application developed during **CCR-001** and **CCR-002**. **Do not create a new program.** Continue building upon your existing codebase.

Business Requirements

The Registrar would like the program to continue performing **all functionality from CCR-001 and CCR-002**.

In addition, the software should:

1. Validate the student's GPA before determining academic standing.
2. Reject GPAs that are outside the valid range.
3. Determine the student's academic standing based on the GPA.
4. Display the student's academic standing as part of the student report.

For this release, use the following academic standing policy:

GPA	Academic Standing
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3.50 – 4.00	Honors
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2.00 – 3.49	Good Standing
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1.00 – 1.99	Academic Probation
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0.00 – 0.99	Academic Suspension
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If the GPA entered is less than **0.00** or greater than **4.00**, the program should display an error message and should **not** determine an academic standing.

Acceptance Criteria

The project will be considered complete when all of the following requirements have been met.

- ☐ The program compiles without errors or warnings.
- ☐ All functionality from **CCR-001** continues to work correctly.
- ☐ All functionality from **CCR-002** continues to work correctly.
- ☐ The GPA is validated before determining academic standing.
- ☐ Invalid GPAs produce an understandable error message.
- ☐ Honors students are classified correctly.

- ☐ Students in Good Standing are classified correctly.
 - ☐ Students on Academic Probation are classified correctly.
 - ☐ Students on Academic Suspension are classified correctly.
 - ☐ The academic standing is displayed as part of the student report.
 - ☐ Output is neatly formatted and professional.
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Technical Constraints

For this week's assignment, you may **only** use C language features covered through **Lesson 8**.

Permitted concepts include:

- Variables
- Constants
- Arrays
- Enumerations (enum)
- User input
- Formatted output
- Mathematical operators
- if statements
- switch statements (where appropriate)
- Logical operators (&&, ||, !)

Do **not** use language features that have not yet been introduced in class.

Out of Scope

The Registrar is **not** requesting any of the following features at this time.

- Multiple students
- Functions
- Files
- Menus
- Editing grades
- Automatic GPA calculation
- Degree audits
- Student transcripts
- Dynamic memory allocation

These features will be requested in future Client Change Requests.

Questions for the Customer

Before writing any code, identify questions you would ask the customer to clarify the requirements.

Examples include:

1. Are the GPA ranges inclusive of the upper and lower limits?
2. Can a GPA contain more than two decimal places?
3. Should the program continue processing after an invalid GPA is entered?
4. Should the user be allowed to re-enter an invalid GPA?
5. What message should be displayed for an invalid GPA?
6. Should the academic standing be stored or only displayed?
7. Should the GPA always be entered manually, or will it eventually be calculated from course grades?
8. Should Honors students receive additional recognition in the report?

9. Will future versions support different academic standing policies for graduate students?
 10. Should academic standing be represented internally using descriptive text or predefined categories?
 11. Should the report include both the GPA and the academic standing?
 12. Should academic standing be recalculated every time the GPA changes?
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Example Program Execution

Example 1

User Input

Student ID: 1234567

Student Name: Alice Johnson

Current GPA: 3.84

Course Grades

95

88

91

84

100

Expected Output

Student Information Management System

Version 3.0

Student Summary

Student ID : 1234567

Student Name : Alice Johnson

Current GPA : 3.84

Academic Standing : Honors

Course Grades

Course 1 : 95

Course 2 : 88

Course 3 : 91

Course 4 : 84

Course 5 : 100

Average Grade : 91.60

Highest Grade : 100

Lowest Grade : 84

Example 2

User Input

Current GPA: 2.45

Expected Output

Academic Standing : Good Standing

Example 3

User Input

Current GPA: 1.52

Expected Output

Academic Standing : Academic Probation

Example 4

User Input

Current GPA: 0.62

Expected Output

Academic Standing : Academic Suspension

Example 5

User Input

Current GPA: 4.35

Expected Output

ERROR

Invalid GPA entered.

GPA must be between 0.00 and 4.00.

Regression Testing Requirements

Because this project is developed incrementally, **all previously implemented functionality must continue to work correctly.**

Your submission must demonstrate that:

- CCR-001 functionality still works.
- CCR-002 functionality still works.
- CCR-003 functionality works correctly.

At a minimum, include the following regression test cases.

Test GPA Expected Standing

- | | | |
|---|-------|---------------------|
| 1 | 3.95 | Honors |
| 2 | 3.20 | Good Standing |
| 3 | 1.75 | Academic Probation |
| 4 | 0.60 | Academic Suspension |
| 5 | -0.50 | Invalid GPA |
| 6 | 4.25 | Invalid GPA |

For **each** test case include:

- Input entered
- Expected output
- Actual output
- Pass/Fail result

If a defect is discovered during testing:

1. Document the defect.
 2. Correct the program.
 3. Repeat the test.
 4. Demonstrate that the correction did not break any previously working functionality.
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AI Usage Policy

Artificial Intelligence tools (such as ChatGPT, GitHub Copilot, Claude, or Gemini) may be used during this assignment.

However, you are responsible for:

- Understanding all submitted code.
- Testing every feature.
- Correcting any defects.
- Ensuring that previously implemented functionality continues to operate correctly.

Submitting AI-generated code that does not function correctly will not receive full credit.

Submission Requirements

Submit a single ZIP file containing:

- main.c
- Any additional source files (if applicable)

Also include a PDF containing:

- Screenshots of all required regression tests
 - A brief description of any bugs discovered
 - A description of how each bug was corrected
 - Answers to the **Questions for the Customer**
 - Any assumptions you made because the customer requirements were ambiguous
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Looking Ahead

This is the **third** release of the Student Information Management System (SIMS).

Future Client Change Requests will continue expanding this application by introducing functions, file processing, pointers, dynamic memory allocation, and additional software engineering concepts.

Remember that professional software developers rarely build software all at once. Instead, they enhance an existing system by implementing new customer requests while ensuring that previously delivered functionality continues to work correctly. Your regression testing should demonstrate that your software remains reliable as it evolves.